A Review on Environment-Adaptive Gait Planning for Semiautonomous Lower Limb Exoskeletons

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***Abstract*—Recent years have witnessed a growing interest in lower limb exoskeletons (LLEs) designed to restore walking capabilities for individuals with motor impairments. Currently, the research community faces the challenge of enhancing the translational impact of this technology by proposing solutions to move from laboratory and clinical settings to daily life scenarios. In this context, the topic of environment-adaptive gait planning (EAGP) is of particular interest and importance. EAGP refers to the ability of LLEs to assist in walking on various terrains (e.g., stairs and slopes) and under different conditions (e.g., the presence of obstacles and uneven ground) by adapting their behavior according to the context. However, the proliferation of publications has led to a general lack of convergence within the scientific community toward a common definition of this research topic. Therefore, this article proposes a review of EAGP for LLEs. The objectives are to: 1) provide a clear definition of EAGP, highlighting its main components and characteristics; 2) propose a taxonomy for classifying the most frequently used techniques in the state of the art; and 3) conduct a comprehensive survey of the literature, situating each article within the proposed taxonomy. In addition, we identify the most urgent future challenges toward the development of more autonomous LLEs.**

***Index Terms*—Assistive technology, computer vision, exoskeleton, gait planning.**

I. Introduction

A POWERED lower limb exoskeleton (LLE) is a robotic system that is worn on all or part of the lower limbs of the user. These devices provide both structural support and power to help perform locomotion tasks. The power source of these devices can vary, but they usually rely on electric motors, pneumatics, levers, hydraulics, or a combination of these technologies [1]. Although LLEs are showing emerging applications in the military [2] and industry [3], robotic exoskeletons are used mainly for the rehabilitation of people with severe gait impairments (e.g., stroke survivors and spinal cord injury patients) [4], [5]. In this context, the first exoskeleton devices were connected to a body weight support system to provide patients with gait training on a treadmill; well-known examples of these systems are the Lokomat [6], ALEX [7], and LOPES [8] exoskeletons. Further developments led to the spread of overground exoskeletons that allow gait rehabilitation in more ecological settings and provide stimulation by navigating the environment [9]. The latest frontier in this field is to translate exoskeletons from rehabilitative systems to assistive devices to ensure patient mobility in everyday settings. However, despite great advances in terms of control, wearability, and usability [10], the uptake of LLEs outside clinical and research environments is still extremely limited [11], [12]. In fact, current assistive LLEs are "nonautonomous" systems in the sense that their motion is still based on guiding the legs through preprogrammed physiological gait patterns, where the parameters can be mildly customized for each patient (e.g., in terms of step height, length, and duration), but remain constant throughout locomotion [13], [14], [15], [16], [17]. To overcome this limitation, the literature on LLEs is focusing on solutions to fully or partially modify gait patterns during locomotion—e.g., to adapt to different terrains such as level ground, stairs, and slopes [18], [19]. In these solutions, the device is generally unaware of the environment in which walking occurs, and the switch between a predefined set of locomotion modes is generally done by means of manual user commands (e.g., push buttons) [20], possibly increasing the user's workload. Thus, a further recent advancement in this direction is the investigation of methods to achieve an automatic transition or generation of locomotion patterns based on environmental perception when user intent alone is insufficient [21], a task that we refer to as environment-adaptive gait planning (EAGP). The implementation of these new approaches represents a fundamental step in fostering the use of LLEs in unconstrained environments by envisioning them not simply as a mere end-effector of the user's commands, but rather as a semiautonomous system capable of contextualizing and complementing the user's commands to optimize its behavior according to the environ-

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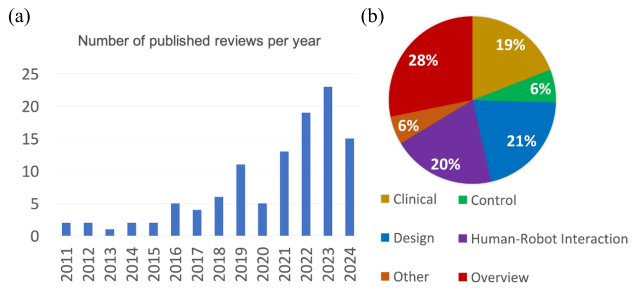


Fig. 1. (a) Bar chart displaying the number of reviews on LLEs published per year from 2011 to 2024. (b) Pie chart displaying the percentage of reviews on LLEs categorized according to the main aim of the review.

ment. Despite the great number of reviews on LLEs and the proliferation of publications on these topics, the literature is still characterized by strong heterogeneity in both the methodology and terminology, leading to a lack of convergence toward a common definition of EAGP. Therefore, this article aims to shed light on EAGP in LLE applications. In particular, we first provide a clearer definition of EAGP inserted in the control framework of traditional LLEs; second, we propose a taxonomy of EAGP methods at the state of the art, providing a comprehensive overview of their application in LLEs. Finally, current challenges and future directions on EAGP in LLEs are discussed.

II. Related Work

The promises and increasing interest of the research community in the development of LLEs are clearly highlighted by the tremendous increase in reviews of the literature on the topic, as shown in Fig. 1(a). By querying the Scopus database with the query TITLE-ABS-KEY (lower AND limb AND exoskeleton AND review), we found that 280 reviews have been published up to January 2025. By focusing on the ones published after 2010 in the field of "Engineering and Computer Science," we restricted the results to 174. After screening out-of-topic results (e.g., reviews focused on passive exoskeletons/orthoses), we ended up with 111 results. We created the following categorization for the resulting reviews, as can be seen in Fig. 1(b): overview (which gives a general and broad description of the topic) [22], [23], [24], design (which includes materials, sensors, and actuation) [20], [25], [26], [27], clinical (which evaluates the effectiveness of current exoskeletons in rehabilitation) [28], [29], [30], control (which focuses on mid- or low-level control strategies) [18], [31], [32], human–robot interaction (including human–machine interfaces) [33], [34], [35], and others (bibliometric analysis [36] and socioeconomic impact of exoskeleton devices [37]). The outcome of this systematic search for LLE reviews showed that most of these reviews focus on design [20], [22], actuation [23], [25], sensing [24], [26], and control [18], [31]. For example, an extensive review of sensor signals (e.g., force and torque sensors, inertial sensors, electromyography, and electroencephalography) and their use in different types of LLE control is provided in [24], while design and actuation choices for different types of LLE are discussed in [20]. In general, the problem related to the generation and adaptation of gait trajectories is not yet well-defined; furthermore, many of these reviews highlight the lack of convergence of LLE research to a common framework and validation protocol [24], [28], [38]. Moreover, the integration of vision in LLE control seems to be overlooked in the literature. An exception is a recent review [39]; however, this article does not explore the link between vision and adaptive gait generation. To the best of authors' knowledge, no review specifically focusing on computational methods and algorithms for EAGP in LLEs is present in the literature although a couple of recent reviews briefly touched on the matter [39], [40]. In general, these publications are evidence of the growing interest in the subject and highlight the need for a more in-depth analysis of EAGP solutions for LLEs. Thus, this article differs from the previous ones by focusing on the algorithmic aspects of gait planning for LLEs. Nevertheless, we provide a complete list of reviews on LLEs in the Supplementary Materials (Table S-I) that readers could refer to, regarding other related topics such as design, actuation, and control hardware of LLEs.

III. Definition of EAGP in LLE Control

According to several reviews on LLEs for rehabilitation and assistance [31], [41], [42], most of these devices are characterized by a hierarchical control strategy composed of three levels, as shown in Fig. 2, with each level responsible for performing different tasks: the high-level control is generally in charge of the human–exoskeleton interface. For example, the high-level controller captures the movement intent via input devices (e.g., push buttons) and neurophysiological signals (electroencephalography and electromyography), and allows the user to manually select, e.g., through control panels, specific device behaviors, such as different locomotion modes (normal walking, running, and stairs ascending/descending). Additionally, at this level, the system acquires and processes data related to the physical human–robot interaction (pHRI), typically measured by force/torque sensors at exoskeleton attachment points. Once the user's commands are recognized, the mid-level control is responsible for managing the behavior of the LLE. To allow a safe control of the system, a finite state machine (FSM) is often employed to translate the user's input into a desired device state (e.g., stand, sit, right step, and left step) and generate the corresponding kinematic models and reference gait trajectories; these trajectories are then converted into appropriate control commands that are sent to the low-level control. This layer is closest to the device actuators and generally implements standard algorithms [e.g., proportional-integral-derivative (PID) controllers] to achieve feed-forward or feedback control, i.e., to account for the kinematic and kinetic characteristics of the actuation such as mass, friction, and inertia. In some cases, a closed-loop control is also implemented by computing the error between the current and the desired gait trajectory to correct the exoskeleton's behavior. Given the state of the art of LLEs, here, we define EAGP as an extension of the general LLE control scheme shown in Fig. 2(a). More specifically, we define EAGP as the generation of gait trajectories that explicitly consider the features of

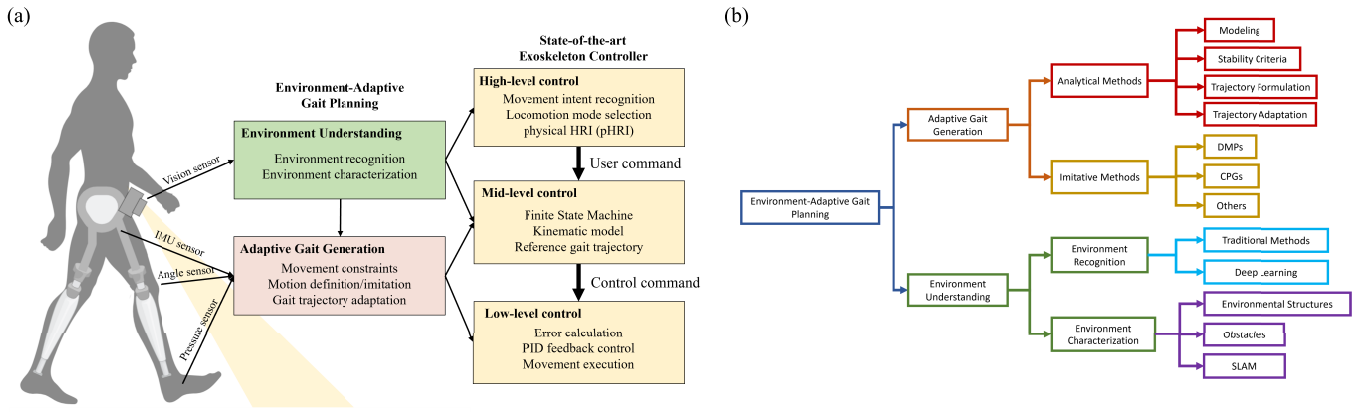


Fig. 2. (a) Schematic representation of EAGP within the state-of-the-art control framework for LLEs. The environment understanding module acquires and processes the data from robotic vision sensors (e.g., LIDAR and camera) to recognize the environment and identify potential obstacles. The adaptive gait generation module then generates at runtime the most appropriate gait trajectory for different locomotion modes and/or to avoid detected obstacles. (b) Proposed decomposition of the EAGP task into adaptive gait generation and environment understanding methods.

the surrounding environment, acquired through vision sensors such as laser range finders and cameras. Given this definition, we can see EAGP as composed of two main components: *environment understanding* and *adaptive gait generation*. Environment understanding is deeply linked with the field of computer vision and refers to the ability of reasoning upon the environmental perception (e.g., acquired through cameras) to produce an accurate representation of the environment. This is achieved through the combination of two main tasks: 1) the automatic identification of the terrain (*environment recognition*) to be combined with the information from the high-level control for the locomotion mode identification, or directly to the mid-level control for the selection of the kinematic model and reference gait trajectory and 2) the detection and localization (i.e., geometry and position) of nearby obstacles and notable environmental structures (e.g., stair steps and slopes) interfering with the normal walking path (*environment characterization*). Adaptive gait generation instead refers to the ability of the exoskeleton to generate and adapt the gait trajectory to different movement constraints (e.g., foothold position and way points) and produce different locomotion patterns. Movement constraints can be derived from exoskeleton internal sensors, such as angle sensors, foot pressure sensors, and inertial measurement unit (IMU), or from vision information from the environment understanding module. The generated trajectories are then sent either to the mid-level control to replace the predefined reference trajectories of the LLE, or converted into control commands (e.g., torque profiles) for the low-level control. In the following, a comprehensive survey of the most widely used methods for adaptive gait generation and environment understanding is reported, organized according to the taxonomy shown in Fig. 2(b). Finally, we report and discuss the works that combine adaptive gait generation and environment understanding into an EAGP solution for LLE applications.

IV. Adaptive Gait Generation

Based on the definitions of motion planning [43] and gait [44], gait generation for biped robots refers to the process of determining the appropriate leg movements and body adjustments that allow a bipedal robot to move from one location to another in a stable and efficient manner [45]. This involves generating a series of joint trajectories that allow the robot to maintain balance, prevent falling, and achieve the desired goal position. This task has been thoroughly studied, as every clinically or commercially available LLE must contain a gait generation module to operate [31]. The process of *adaptive gait generation* sets further constraints on the definition of gait generation, explicitly requiring the ability to modify the gait at runtime to generate different motion patterns for different scenarios. To overcome the aforementioned lack of convergence to a common framework, herein, we propose the following classification of adaptive gait generation methods: *analytical methods* and *imitative methods*. The analytical methods make use of hand-crafted models and equations to preprogram the desired gait trajectories. These trajectories can then be fine-tuned to comply with various constraints (e.g., different terrains). On the other hand, imitative methods consist of data-driven approaches aimed at imitating physiological gait patterns through the learning of prerecorded reference trajectories.

A. Analytical Methods for Adaptive Gait Generation

Historically, analytical methods have been the first viable solution for gait planning, since data-driven methods (e.g., machine learning applied to robot locomotion) were not prominent until recently [43], [46]. The general control scheme that characterizes the analytical methods for LLE gait generation is shown in Fig. 3. Analytical methods comprise four main elements:

1) *dynamic model*, which describes the expected dynamics of the LLE;
2) *stability criteria*, which express the necessary conditions for dynamic balance based on the model;
3) *trajectory design*, which determines the shape of the basic trajectories describing the gait kinematics;
4) *trajectory adaptation*, which enforces motion constraints on the basic trajectories in order to adapt the gait, e.g., based on real-time sensor measurements.

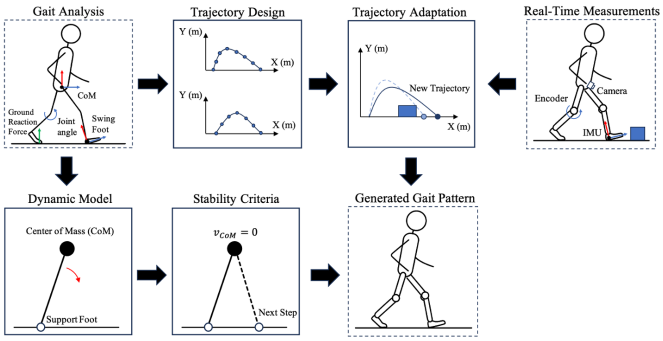


Fig. 3. Main elements of analytical methods. The dynamic model and trajectories are extrapolated from gait analysis; their parameters are then adapted based on the current scenario in order to produce the final trajectories.

In the following, the most adopted solutions for each of these four main elements are described.

1) Dynamic Model: The appropriate choice of a mathematical model is fundamental for describing and controlling the behavior of a robotic system [47]. The most general and comprehensive model is that of a multilink robot, where each link is described in terms of geometrical (e.g., dimensions) and mechanical (e.g., mass and inertia) parameters [48]. Such a detailed model requires an exact description of the mechanical properties of the overall system to achieve high control performance; however, this leads to very complex dynamic equations [49]. Consequently, multilink models are rarely used for adaptive trajectory generation in LLEs. Instead, simpler models are preferred. In particular, the most notable one is the inverted pendulum model. In fact, the inverted pendulum (and its derivations) has become by far the most widely used model for LLE applications [19], [50], [51], [52], [53], [54]. In this model, the bipedal walking motion is approximated by assuming that the center of mass (CoM) of the LLE (typically at hip level) acts as the pendulum bob, the support leg is the pendulum rod, and the center of pressure (CoP) of the support foot is taken as a fixed point. These assumptions greatly decrease the number of model variables while still capturing the fundamental elements of bipedal locomotion. Moreover, derivations of this model—such as the linear inverted pendulum model (LIPM) [55], [56], [57], [58] and the linear inverted pendulum flywheel model [59] (which uses a flywheel to approximate the hip-trunk connection and account for rotational forces)—are also found in the literature.

2) Stability Criteria: Once a dynamic model is selected, it is necessary to impose stability criteria to ensure dynamic balance during gait. Although external supports (such as crutches) can reduce the risk of falling, loss of balance remains a critical factor to address. The zero moment point (ZMP) criterion, developed by Vukobratovic and Borovac [60], is the most notable stability criterion in LLE applications. The ZMP is defined as the point on the ground at which the horizontal components of the reaction forces do not produce any net moment. In a multilink robot, the ZMP is given by [61], [62]

$$x_{\text{zmp}} = \frac{\sum_{i=1}^{n} m_i\left(\ddot{z}_i + g\right) x_i - \sum_{i=1}^{n} m_i \ddot{x}_i z_i - \sum_{i=1}^{n} I_{\text{iy}} \ddot{\theta}_{\text{iy}}}{\sum_{i=1}^{n} m_i\left(\ddot{z}_i + g\right)} \tag{1}$$

$$y_{\text{zmp}} = \frac{\sum_{i=1}^{n} m_i\left(\ddot{z}_i + g\right) y_i - \sum_{i=1}^{n} m_i \ddot{y}_i z_i - \sum_{i=1}^{n} I_{\text{ix}} \ddot{\theta}_{\text{ix}}}{\sum_{i=1}^{n} m_i\left(\ddot{z}_i + g\right)} \tag{2}$$

where x_i, y_i, and z_i are the CoM coordinates for each link, and I_{iy} and I_{ix} with $\ddot{\theta}_{iy}$ and $\ddot{\theta}_{ix}$ are the corresponding inertial terms. When applying the ZMP to the inverted pendulum model, especially under the LIPM assumptions (i.e., the CoM moves in a straight line at a constant height z_c), the equations simplify to

$$x_{\text{zmp}} = x_c - \frac{z_c}{g}\ddot{x} \tag{3}$$

$$y_{\text{zmp}} = y_c - \frac{z_c}{g}\ddot{y}. \tag{4}$$

Here, (x_c, y_c) is the CoM position in the horizontal plane. In conjunction with the inverted pendulum model, the ZMP formulation (and other stability criteria such as capture point [57], [58], [59], extrapolated CoM (XCoM) [63], or the divergent component of motion (DCM) [64]) serves to define a clear relationship between the CoM dynamics and the support foot dynamics. Such formulations are often structured to allow real-time adaptation by estimating the model parameters and generating the desired system behaviors (e.g., controlling the CoM to obtain a desired ZMP trajectory, as in [65]).

3) Trajectory Design: The key aspect characterizing analytical methods for adaptive gait generation is that gait trajectories are designed through hand-crafted equations, whose formulation is typically derived from the movement analysis of human locomotion [66]. In general, differentiable functions are considered in order to customize the trajectory not only in position, but also in velocity and acceleration. Polynomial interpolation, whether it is by using a single polynomial or in a piecewise fashion (splines), is the most common choice in LLEs for the design of the gait trajectory. In fact, by increasing the degree of the polynomial, it is easy to model more complex trajectory behaviors. Cubic polynomials allow for fast computation although they can only account for trajectory position and velocity [67], [68], [69], while quintic polynomials can lead to precise control over acceleration [59], [70], [71]. Higher order polynomials grant additional flexibility, such as control over the curvature of the trajectory [72], [73], [74]. Another common approach to trajectory design is the use of sinusoidal functions [52], [54], [55], [56], [75], [76], [77], thus a weighted sum of (hyperbolic) sine and cosine functions. In this case, the parameters to be determined are the amplitude, frequency, and phase of each sinusoidal component. Although this type of trajectory can also be applied to the swing foot, it is commonly used to approximate the CoM trajectory: since inverted pendulum models provide a CoM trajectory which can be expressed as a sum of exponential functions, hyperbolic functions give an equivalent representation which is usually employed in these models [54], [76].

4) Trajectory Adaptation: The trajectory design provides a parameterized template that describes the gait kinematic (e.g., the swing foot trajectory), with an initial value for the parameters. In order to adapt the trajectory template, some motion constraints must be defined to compute the actual parameters of the trajectory function for the specific scenario.

The simplest type of trajectory adaptation is obtained through the variation of general descriptive characteristics of the gait, such as step length, height, and duration: in these cases, the adaptation is obtained through a spatial or temporal scaling of the reference trajectory without modifying its shape [74], [78]. More complex motion constraints can be defined to have a more flexible adaptation of the full gait trajectory. This generally includes the definition of specific way points for the trajectory, such as starting, intermediate, and ending positions of the swing foot [69], [79]; thanks to the use of differentiable functions, velocity, acceleration, and higher derivative constraints can be posed to ensure smoothness and avoid sudden changes in position [72], [73]. Kinematic constraints can also be applied to ensure the feasibility of the generated trajectory (e.g., the hip-to-foot distance must not exceed the leg length, or the maximum trajectory angles do not exceed the range of motion of the exoskeleton joints) [79]. The process of translating the trajectory constraints into the trajectory function parameters usually involves hand-crafted mathematical relationships that produce an optimization problem. Most of them result in closed-form expressions that can be solved through direct interpolation [54], [76], [78], or through more sophisticated methods such as Hamiltonian function optimization [19], [50], [51], [80]. When there is a high number of parameters to optimize, nonlinear optimization methods such as nonlinear programming [81] or nonlinear model predictive control (MPC) have also been employed [82]. Other works make use of stochastic optimization [68], [69] to iteratively identify a suitable gait trajectory that satisfies the constraints (e.g., avoiding foot collision with an obstacle). More recently, reinforcement learning (RL) has also been explored in the context of adaptive gait generation for LLEs: in this case, the trajectory is produced by the sequence of actions of the agent and the motion constraints are defined in the RL reward function [79], [83].

B. Imitative Methods for Adaptive Gait Generation

Imitative methods are data-driven approaches aimed at generating a human-like gait; differently from analytical methods, the key aspect of imitative methods is that a data collection phase is necessary to learn reference gait trajectories directly from human demonstrations. Machine learning techniques are then generally used to reproduce and adapt the desired gait trajectories at runtime, e.g., based on data from robotic sensors such as exoskeleton joints' encoder or force/torque measurement. Based on recent literature, two main techniques stand out in terms of frequency of usage and results: dynamic movement primitives (DMPs) and central pattern generators (CPGs).

1) Dynamic Movement Primitives: DMPs were first proposed by Schaal [84] and later updated by Ijspeert et al. [85]. They provide a framework for encoding elementary units of movement as stable nonlinear attractor systems, where the goal corresponds to the desired pose at the end of the movement. DMPs can learn arbitrary motions (periodic or aperiodic) by adding a nonlinear term to a dynamical system, representing the behavior of a single degree of freedom (DoF); by applying machine learning techniques on the nonlinear component, different motions can be generated. Moreover, different DoFs can be coordinated by adding coupling terms that represent the relationship between the movement of multiple joints [85]. DMPs have been extensively applied in autonomous robotics because they can generate complex rhythmic (e.g., locomotion) and discrete (e.g., tennis swing) behaviors while quickly adapting to perturbations in a dynamically changing, stochastic environment [86], [87]. In the general framework proposed by Ijspeert et al. [85], three main ingredients constitute a DMP (displayed inside the red box in Fig. 4).

1) *Canonical system* is a simple dynamical system that generates a phase variable to replace the explicit time variable. This phase variable is normalized between 0 and 1, yielding an invariant representation of time and facilitating the generalization of both discrete and periodic behaviors. For example, a first-order linear system is commonly chosen as the canonical system for discrete movements [57], [85], while for rhythmic motions, one may use phase oscillators or even Hopf or van der Pol oscillators [88].
2) *Nonlinear Function Approximator:* Taking as input the phase information from the canonical system, it generates a forcing term that customizes the trajectory toward the desired reference, typically learned offline from demonstrations via a machine learning algorithm.
3) *Transformation System:* It defines another dynamical system that produces the desired motion when modulated by an external forcing term (e.g., real-time measurements from the robotic sensors). The choice of the transformation system is often dictated by the requirements of the low-level controller: if, for example, the system is controlled in acceleration, then the transformation system must provide smooth second derivatives of the position trajectory.

As an example, a common formalization of a discrete DMP system for a single DoF (e.g., a single robot joint) is given by

$$\tau \dot{z} = \alpha_z \left(\beta_z \left(g - y\right) - z\right) + f, \qquad \tau \dot{y} = z \tag{5}$$

$$\tau \dot{x} = -\alpha_x x \tag{6}$$

where f is a nonlinear function composed of weighted basis functions

$$f\left(x\right) = \frac{\sum_{i=1}^{N} \psi_i\left(x\right) w_i}{\sum_{i=1}^{N} \psi_i\left(x\right)} \, x \, \left(g - y_0\right) \tag{7}$$

$$\psi_i\left(x\right) = \exp\left(-\frac{1}{2\sigma_i^2}\left(x - c_i\right)^2\right). \tag{8}$$

In the example of (5), the transformation system is formulated as a spring–damper model, where g represents the goal state (e.g., the target joint angle), α_z and β_z are the positive constants, and τ is a temporal scaling factor. Here, y and $\dot{y}$ correspond to the desired position and velocity, respectively. Equation (6) describes the canonical system that produces the phase variable x, with α_x a constant. Finally, (7) and (8) describe the nonlinear function approximator, where y_0 is the initial position, and σ_i and c_i determine the widths and centers of the basis functions ψ_i, respectively. The versatility of the DMP framework has led to numerous extensions and

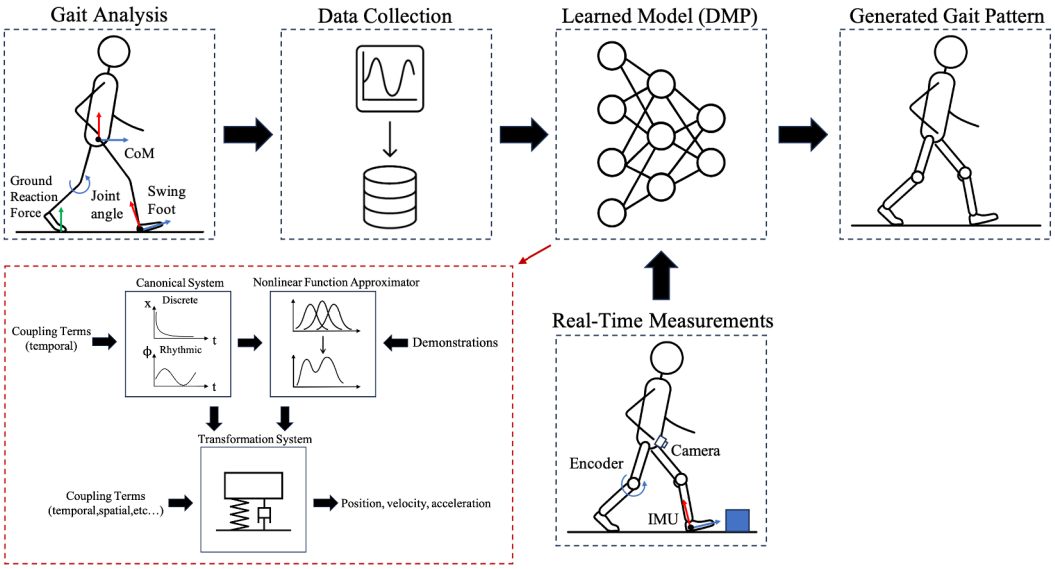


Fig. 4. General design of DMP-based methods: after data are collected from gait analysis (with the possible addition of other data sources, e.g., environmental), the learned model can generate adaptable trajectories based on the received input. The red box represents the principal elements of the DMP learning framework: canonical system, nonlinear function approximator, and transformation system.

applications in LLE gait planning; a scheme comprising the most important element of DMP-based solutions for LLEs is shown in Fig. 4. For example, a DMP-based approach that uses a first-order canonical system coupled with locally weighted regression (LWR) to learn the nonlinear forcing term f has been used to adapt the swing foot trajectory of the exoskeleton for uphill slopes [57], [58], [89]: the nonlinear function approximator first learns the reference trajectory from level-ground walking demonstrations. Then, new trajectories are generated by varying starting and ending position constraints to account for different slope angles. Furthermore, the DMP framework has been extended to tasks such as ascending stairs with good performance [90], [91], [92]. Beyond LWR, alternative machine learning techniques have been explored to learn the parameters of the nonlinear function. These include Gaussian mixture regression [64], Gaussian regression neural networks [93], and even RL approaches [91], [94]. Another interesting application is represented by Chen et al. [95] and Luo et al. [96], in which DMPs are not used to learn gait patterns for different terrains; rather, to obtain a more accurate execution of the gait trajectory in level-ground walking by treating the difference between the actual and desired gait trajectory as a forcing term. In particular, in [96], the reference swing foot trajectory is provided in the joint (angular) space, thereby obviating the need for inverse kinematics and providing a more direct input to the exoskeleton low-level controller. Variants of the original DMP formulation have also been proposed. Stylistic movement primitives (SDMPs) [97] introduce a style parameter that allows the generation of different motions from multiple reference trajectories and have been applied in LLE research to learn the gait trajectory at different velocities [98]. Another variant is represented by coupled cooperative primitives (CCPs) [99], in which the standard first-order linear system, which usually describes locomotion, is coupled to an impedance spring–damper model that accounts for the pHRI. The CCP model parameters are first learned offline from a gait analysis dataset and then adapted online through the PI^2 RL algorithm [89]. In addition, kernelized movement primitives (KMPs) [100] replace standard basis functions with generic kernels learned through Gaussian mixture regression, and have been used in exoskeleton application to adapt gait to different walking speeds [101] or to different terrains such as slopes and stairs [102], [103]. All the previously described works are examples of DMP methods based on transformation systems meant to learn discrete movements (e.g., the swing foot trajectory during a single step). A notable exception in the literature is provided by Huang et al. [104] and Wang et al. [105], where rhythmic DMPs are used to generate angular knee trajectories in a single DoF LLE. In these works, a limit-cycle oscillator is employed as dynamical system (Hopf oscillator [104] and Von-Mises basis functions for rhythmic movements [105]) to learn the full gait cycle, whose parameters are learned offline through LWR from reference knee trajectories; by measuring the difference between the reference and actual joint angles, the DMP model adjusts the output trajectory in real time. In particular, in [105], another variant to the classical deterministic DMPs is proposed, namely, the probabilistic movement primitives (ProMPs) [106]. ProMPs use probability distributions to encode the movement that allow to incorporate the variance of the movement from multiple demonstrations, similar to KMP [100].

2) Central Pattern Generators: CPGs are neural networks capable of producing coordinated rhythmic patterns of activity from nonrhythmic inputs (e.g., sensory feedback or signals from higher control centers) and form the basis for many rhythmic behaviors in both invertebrates and vertebrates [107], [108]. Physiologically, the vertebrate locomotor system is organized such that spinal CPGs generate the basic rhythmic patterns, while higher level centers (e.g., motor cortex, cerebellum, and basal ganglia) modulate these patterns according

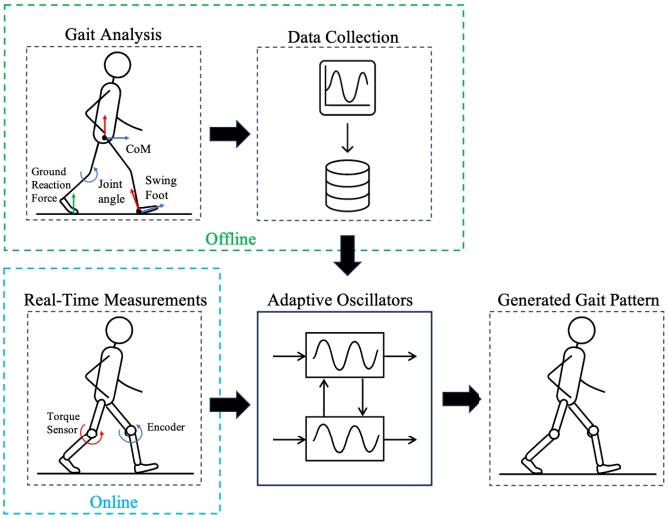


Fig. 5. General template of CPG-based solutions for LLE. The two main categories (offline and online) are united by the use of adaptive oscillator networks to produce periodic trajectories.

to environmental conditions [109], [110]. Normally, a CPG unit controls one DoF, and a CPG network coordinates a set of joints to complete a movement. In short, by modulating the parameters of a CPG network, output signals with different phase relationships can be generated; these phase relationships can be used to obtain different gaits [108]. Considering the general formulation of CPG, different CPG models have been proposed at several levels of abstraction depending on the phenomenon under study, from detailed biophysical models to connectionist models and abstract systems of coupled oscillators [107]. Among these, oscillator-based and biophysical CPG models are the most common implementations for bipedal locomotion [108]. The general design scheme of CPG methods for LLE application is depicted in Fig. 5. In oscillator network models, input can be provided as a rhythmic reference trajectory or a high-level performance criterion (for example, move as fast as possible), while biophysical models rely on real-time sensory feedback from each joint (such as forces or torques applied by the exoskeleton) to modulate network output [108]. In both cases, the oscillator parameters—frequency, amplitude, and phase—are tuned through machine learning techniques. For instance, when a reference trajectory is available (e.g., for the swing foot or joint angles), supervised learning methods can be used to identify the CPG network parameters that minimize an explicit error function (such as the mean absolute error between the reference and the network output) [111], [112]; in this common case, a supervised CPG network can be composed of coupled adaptive oscillators (such as Hopf oscillators), where each joint is assigned an oscillator that learns a different frequency component from the reference trajectory, analogous to a Fourier decomposition. In [113], for example, a supervised adaptive Hopf (AHopf) oscillator CPG network with multiple oscillators per joint (five for the hip joint and two for the knee joint) was designed to control a 4-DoF exoskeleton by imitating prerecorded periodic angular trajectories. A frequency update law was also introduced to reset the oscillator phases based on heel strikes, adapting the output to the current step frequency. Supervised AHopf oscillators have been similarly applied to generate Cartesian trajectories of the swing foot for hemiplegic users in a 4-DoF LLE [114]; in this case, the trajectories of the healthy leg serve as reference, and an improved differential evolution (iDE) algorithm [115] searches for optimal oscillator parameters. Interestingly, Akkawutvanich and Manoonpong [116] propose an implementation that decouples the problem into separate learning modules. In particular, a supervised neural network-based approach separates rhythmic pattern generation (using decoupled neural oscillator networks with Hebbian synaptic plasticity [111]) from a postprocessing module based on radial basis function (RBF) networks. In this setup, the RBF networks translate the rhythmic output from the first network into the joint angular trajectories of the single step, learned from reference trajectories (for hip, knee, and ankle joints) recorded with the exoskeleton in transparent mode (i.e., compensating only for its mass, friction, and inertia). When no specific reference trajectory is defined, the other family of CPG approaches is based on unsupervised learning techniques. In this scenario, the CPG network receives one or more afferent signals (e.g., current joint positions and velocities) and learns the desired motion by optimizing criteria such as desired walking speed or CoM stability. A particularly common input in the context of LLEs is the sensors' signals from the pHRI system [117], [118], [119]. For example, in [117], unsupervised CPGs based on nonlinear coupled oscillators (with a single oscillator per joint and coupling between adjacent joints reflecting biomechanical constraints) are implemented. This network, designed to take the human–exoskeleton interaction torques as input [estimated through a nonlinear autoregressive exogenous (NARX) network fed with current joint trajectories and velocities], adapts the CPG parameters in real time to align with the user's intent, thus generating robust and stable periodic trajectories that alleviate the burden on higher level control. Subsequent work by Mehr et al. [118] improved on this approach by replacing NARX with a deep neural network (DNN) for torque estimation and using a random forest regression model [120] to evaluate the prediction uncertainties. Such measures are crucial because inaccurate torque estimates can lead to erroneous adjustments of CPG parameters and, consequently, unsafe system behavior. In another study [121], deep RL was used to adjust the CPG parameters based on pHRI with the aim of minimizing interaction torques and ensuring better synchronization between the user and the exoskeleton. A less common family of CPG methods in exoskeleton applications is biophysical CPG models based on Matsuoka neurons [122], [123]. In [124], such models were used to generate bioinspired walking gaits for a 6-DoF LLE in conjunction with musculoskeletal analysis. Here, two oscillators per joint (representing antagonist muscles) are employed, with sensory feedback (dependent on joint angles) modulating the neuron outputs that, together with foot contact force measurements, determine the joint torques. However, the experimental validation of these biophysical approaches remains in a preliminary stage [124].

3) Other Imitative Methods: Even if DMPs and CPGs currently represent the most used imitative methods for gait

learning and generation, other imitative methods have been studied. Recurrent neural networks (RNNs) are very efficient in sequence data representation and prediction tasks, also including gait data analysis [125]. Among RNNs, a long-short term memory (LSTM) network was applied to skeleton-based action recognition [126], and to learn the behaviors of a humanoid robot based on multiple sequences of joint data for motor behavior recognition [127]. In [128], a deep spatial–temporal model (DSTM) is proposed, which leverages LSTM layers to learn the inherent spatial–temporal correlations of the gait features to generate the trajectory of the knee joint for LLEs. In particular, the proposed DSTM predicts the pathological trajectories of the knee joints and produces a corrected knee trajectory profile based on the evolution of the trajectories of the other joints. A similar use of LSTM for LLEs was proposed in [129]: by developing a gait prediction model to infer future angular trajectories of the lower extremities based on real-time joint angle measurements, more compliant and robust trajectories are generated. The experiments showed promising results, although the solution was tested with non-disabled users only. In [130], an adaptive stair-ascending gait generation algorithm is proposed for LLEs: stair-ascending gait data from healthy people are collected to obtain reference trajectories that are adapted to different foothold positions (e.g., based on the height of the stair step) through linear interpolation. Another approach for controlling the exoskeleton through data collected from healthy individuals is presented in [131] for a gait rehabilitation exoskeleton. The method includes data acquisition and normalization, key event identification (i.e., six key events of the gait cycle are detected from recorded data), and linear interpolation to generate new trajectories from the reference ones. In particular, the six key events serve as reference points for a closed third-order polynomial spline computation that represents the ankle trajectory of the exoskeleton in the sagittal plane. This provides the controller with the ability to adjust gait trajectories to different rehabilitation protocols by only modifying six parameters as key events. A different approach is proposed in [132]: in this case, reference trajectories of joint positions are obtained through torque compensation control, which compensates for gravity, inertia, and friction to obtain gait data from the user while the exoskeleton is transparent. Based on these reference trajectories, linear interpolation and numerical approximation are used to adapt the parametric gait control to different velocity, step length, and step height parameters. Finally, in [53], the principal component analysis (PCA) is used to infer the lower limb trajectories based on prerecorded full-body trajectories; in this way, the natural synergy between the upper extremity and the lower extremity during locomotion is taken into account in the exoskeleton control. This synergy has been considered both at the higher (planning) and lower (control) levels.

V. Environment Understanding

The adaptation of the gait to the environment is highly dependent on the ability of the device to rapidly and accurately understand its surroundings. Vision-based techniques have emerged as a key enabler in this context, allowing the exoskeleton to detect obstacles [78], [82], assess terrain variations [133], and allow the locomotion mode to change in real time [134]. To do so, the first step is to equip the LLE with one or more exteroceptive sensors to perceive the environment: one of the most common and effective sensors for achieving this is the RGB-D camera [82], [135], [136], which offers the dual advantage of capturing both color and depth information. This rich sensory data supports robust scene reconstruction and obstacle detection, providing fundamental information for dynamic adjustment of the gait of the exoskeleton. Other sensors such as RGB cameras [137], [138], [139], laser range finders [140], [141], and radar sensors [142] have also been employed in the literature. In this article, we separate environment understanding methods for LLEs into two main categories: environment recognition and environment characterization. The first category deals with the classification of input from the camera to provide a high-level label of the scene (e.g., stairs and slopes) for automatic selection of locomotion modes [134], [136], [143]. The second category, environment characterization, provides precise geometric information (e.g., location, pose, and shape) about notable environmental structures (e.g., stairs, curbs, and obstacles), enabling the LLE to perform gait adaptation tailored to the specific scenario.

TABLE I
Comparison of Different Environment Recognition Classifiers Derived From [144]

Classifier	Accuracy	Computation Time
Hough Transform with Gabor Filter or Canny Edge Detector	Not Specified	8 seconds
Gabor Barcodes and Hamming Distances	88.5%	0.15 seconds
SURF Features and Bag of Words	86%	Not Specified
Heuristic Thresholding and Edge Detector	98.8%	0.2 seconds
Cubic Kernel SVM	94.1%	14.9 milliseconds
CNN (27 million parameters)	99.6%	50 milliseconds

A. Environment Recognition

Herein, we separate the various contributions to the task of environment recognition into methods employing traditional computer vision and machine learning techniques, methods exploiting the novel advances in deep learning for image classification, and works providing open-source datasets necessary for training the abovementioned supervised learning models.

1) Traditional Methods: Laschowski et al. [144] performed a comparative analysis of environment recognition systems for LLEs with the objective of evaluating the performance of state-of-the-art techniques (Table I). Environment recognition systems for LLEs have traditionally used statistical pattern recognition and machine learning for image classification. These algorithms require meticulous hand-engineering to develop feature extractors (e.g., edge detectors) that transform raw images into suitable representations or feature vectors. Manually extracted features serve as input to image classification (e.g., machine learning or threshold-based algorithms). Hough transforms, along with Canny edge detectors or Gabor filtering, have been used for edge detection. Support vector machines (SVMs) have also been used to classify depth images of human locomotion environments; different dimensionality reductions and SVM model designs were compared in [135] and [136]. The best model achieved high

TABLE II
COMPARISON OF DIFFERENT DATASETS FOR ENVIRONMENT RECOGNITION IN LLES

Author	Sensor	Data type	Environments	Number of instances
Laschowski et. al [139]	GoPro Hero4 Session	RGB	Flat ground, stairs	34,254
Kurbis et. al [145]	iPhone XS Max	RGB	Flat ground, stairs, transitions	515,000
Laschowski et. al [146]	iPhone XS Max	RGB	Flat ground, stairs, wall, door, transitions	922,790
Patil et. al [147]	Logitech HD Camera C920	RGB + Bounding Box	Stairs	1,696
Wang et. al [148]	Not detailed	RGB + Bounding Box	Stairs	3,094
Wang et. al [149]	Realsense D435i	RGB-D + Bounding Box	Stairs	2,996
Wang et. al [150]	Realsense D435i	RGB-D + Bounding Box	Stairs	2,832

image classification accuracy in five activity classes (running, walking, standing, ascending, and descending the stairs) using a cubic kernel SVM and no dimensionality reduction, reaching 99% accuracy for a single-subject dataset [135], and 94.1% for a dataset containing multiple subjects [136]. Although SVM models are effective in high-dimensional space and offer good generalization (i.e., robustness to overfitting), these algorithms require manual selection of kernel functions and statistical features from regions of interest.

2) Deep Learning: Deep learning replaces manually extracted features with multilayer networks that automatically learn optimal image features from training data. In [134], a convolutional neural network (CNN) is proposed, reaching 99.6% test accuracy in recognizing three environments (level ground and up/down stairs). Similarly, Laschowski et al. [143] employed a CNN to classify up to 12 classes (level ground, up/down stairs and ramps, wall/door, and transitions between different environments), achieving 72.6% of accuracy. Recently, Vision Transformers (ViTs) have also been employed in the context of environment recognition for LLEs: in [151], the employed ViT architecture reached 99.6% accuracy over six classes (up/down stairs and ramps, level ground, and a class for all other cases such as obstructions or untraversable terrain). All of these networks are trained following a supervised learning paradigm, which means that they require a high amount of labeled images from different environments to achieve good performance. To overcome this limitation, in [152], a simulation-to-real training approach is proposed to improve recognition performance. CNN is first trained in an augmented simulated environment, and then, unsupervised domain adaptation is used to transfer the network to real image classification, achieving an accuracy of 96.98% over five classes (level ground, up/down stairs, and up/down ramps).

3) Datasets: Several datasets have been created in the context of environment recognition for LLEs, recording RGB [139], [145], [147], [148], and RGB-D images [149], [150] in the most commonly encountered environments. In Table II, we list these datasets reporting the sensor used for data acquisition, the type of data, the recorded environments, and the size of the dataset. Among those, the most comprehensive is the ExoNet database, the largest and most diverse open-source dataset of wearable camera images of indoor and outdoor real-world walking environments [146]. The dataset contains 922 790 images and 12 classes consisting of different environments (i.e., level ground, inclined/declined stairs, wall, and door) and the transitions between them. Many of the aforementioned environment recognition solutions have been trained and tested on this dataset, making it an implicit benchmark tool for this task.

B. Environment Characterization

While the methods presented in Sections V-A provide only a high-level label to the environment in which walking occurs, environment characterization techniques aim at providing the exoskeleton with a geometric description of the main environmental structures on the walking path (e.g., the length and height of stair steps, the tilt of slopes, pose, and shape of obstacles). This information is crucial to generate an appropriate and safe gait pattern, but usually requires extensive processing of the vision input.

1) Environmental Structure Characterization: One of the first approaches to the characterization of stairs for LLEs has been proposed in [153]: in this work, depth images are processed with traditional computer vision techniques, such as directional and depth thresholding, and edge detection. In [154], different environments (level ground, inclined/declined slopes, and stairs) are detected from 3-D point clouds using the random sample consensus (RANSAC) algorithm [155]; the solution has been validated on a proprietary dataset of 9217 depth images, showing promising results. The advantage of this method is that environmental structure measurement can be inferred with high precision (classification accuracy of 88.4% and measurement deviation of 3.66%). In [70], a method for precise ascending stairs measurement is proposed; a region growing algorithm (i.e., K-nearest neighbor coupled with singular value decomposition) is employed to cluster the unstructured point cloud, and each region is then smoothed by an efficient implementation of L0-norm minimization. Similarly, in [130], a staircase model based on a linked list is constructed based on the detection of the tread (horizontal) and riser (vertical) planes. Then, an optimal foothold point is calculated based on the linked-list stairs model. Although RGB-D cameras provide a rich source of information, low-cost range sensors can also be employed with successful results, as done in [72]: in this case, two range sensors are used to gather input points, which are then clustered and postprocessed through recursive total least-squares regression to characterize stair scenarios. Unlike the other methods listed above, in [149], stairs are first recognized and located using a CNN that employs both RGB and depth images; then, a postprocessing based on stair line clustering and coordinate transformation is used to estimate the geometric parameters of the stairs. A similar approach is applied in [156], where, after a CNN-based classification of the environment, the parameters of the five classes (level ground,

up/down stairs, and up/down slopes) are determined by using RANSAC (for stairs) and regression (for slopes).

2) Obstacle Characterization: Obstacle and object detection is a well-known computer vision task for which many solutions have already been devised; a recent article by Zou et al. [157] extensively reviews the most prominent solutions in the last 20 years. As the article points out, deep learning has been shown to outperform other solutions. Obstacle classification and localization for LLEs through the use of deep learning was proposed in [158]; a modified version of the VGG architecture is trained on both RGB and depth images to detect, classify, and localize obstacles. In addition, the top surface of the obstacle is obtained using the RANSAC algorithm. The model achieved a classification accuracy greater than 87% for six different obstacle classes (big/small rock, dustbin, basketball, car, and nothing) with obstacle sizes ranging from 20 to 100 cm; furthermore, it achieved a top surface positioning error of 0.8–1.2 cm for obstacles of different distances and heights. A different approach was proposed in [82]; based on recorded point clouds, obstacle subsets are individually distinguished by using RANSAC and Euclidean clustering. Then, to obtain a compact envelope that fits tightly to the obstacle surface, a hybrid bounding box is designed; this technique also allows to produce suitable bounding boxes for partially occluded obstacles. The use of RANSAC coupled with clustering has also been documented in [78], although in this case, only the obstacle closest to the LLE is evaluated; through the use of geometric calculations applied to the obstacle point cloud, the dimensions of the obstacle can be inferred. Similarly, in [68] and [69], RANSAC is used to separate obstacles from the level ground although, in this case, all obstacles points occluding the stepping path are considered after a geometric thresholding is applied: therefore, there is no separation between different obstacles, but just between the obstacle points that occlude the walking path and those that do not. In [159], eye gaze mapping is proposed to locate obstacles; two cameras are used to track the position of pupils, and a waist-attached depth camera is used to obtain environmental information. The user gaze is mapped to understand which obstacle is detected; when the obstacle is close enough, a transition gait pattern is generated to traverse it.

3) Simultaneous Localization and Mapping: Although obstacle detection is fundamental in achieving obstacle avoidance, another important aspect is often overlooked in LLE applications: localization of previously detected obstacles that are outside the sensor field of view (FOV). For example, when an obstacle is very close to the feet, it may fall outside the FOV of a waist-mounted camera (which is a typical position to place the camera in these applications [144]); nevertheless, when generating the swing foot trajectory, the obstacle must be taken into account to avoid collisions. This problem is commonly solved through simultaneous localization and mapping (SLAM), a task that consists of creating a map of the surrounding environment and locating the robot inside this map at the same time [160]. Although SLAM solutions have not been extensively applied to LLEs, a preliminary use of this technique is presented in [161], where a vision-based environmental perception pipeline for LLEs for obstacle crossing tasks is proposed. The pipeline consists of three stages, namely: 1) ground plane and obstacle detection; 2) estimation of obstacle location and dimensions; and 3) obstacle tracking. To reduce noisy artifacts and reliably detect obstacles, a similarity metric based on color, gradient orientation, and 2-D surface normal is employed. The depth map of the detected obstacle region is utilized to estimate the location and dimensions of the obstacle. In addition, two obstacle tracking modes for obstacle crossing are considered: visual tracking using an RGB-D camera and positional tracking using a SLAM-integrated depth camera (RealSense T265). A further advancement in the integration of SLAM techniques and LLEs has been achieved by Exosense, a visual odometry scene understanding system [133]. In the proposed solution, a multisensor device collects wide-angle RGB images, depth data, and inertial measurements; visual-inertial odometry (openVINS) is employed for high-frequency state estimation and integrated into a semantic pose-graph SLAM framework (LEXIS) that fuses terrain maps with semantic room labels to build globally consistent, reusable scene representations. The system further refines these maps via terrain reconstruction and traversability analysis to support reliable navigation in indoor and multistory environments. Experiments demonstrate that Exosense achieves low odometry drift (about 4 cm per meter traveled) and high mapping accuracy, paving the way for long-term exoskeleton localization and navigation in complex environments.

VI. Environment-Adaptive Gait Planning

In our proposed taxonomy, EAGP consists of solutions that employ environment understanding techniques whose output produces environmental constraints that become the input of the adaptive gait generation. Due to its robustness and fast computation times, the use of RANSAC is prevalent when it comes to plane recognition, be it to identify level ground [69], [78], [130] or to identify risers and treads that make up stairs [130], [162]; an alternative solution is PolyLidar3D [163], used in [161], whose main strength is the low-dimensional representation of flat surfaces such as polygons. When it comes to stairs, after the risers and treads are identified, a connectivity procedure based on the position and orientation of each plane is often employed to link the subsequent planes to produce the final representation of the stairs [130], [162]. Another common technique in EAGP solutions is clustering. In [72], Euclidean clustering is applied to identify risers and treads, representing a valuable alternative to the use of RANSAC for the same task, while other works employed Euclidean [78], [161] and KD-tree clustering [82] to distinguish individual obstacles. A different approach to obstacle characterization is shown in [159], where gaze mapping is used to localize the obstacle; based on the area pointed by the gaze, geometric analysis is applied to infer obstacle parameters. Most of the EAGP solutions for obstacle avoidance employ analytical methods for gait generation: starting from a parametric reference trajectory, different optimization methods are introduced to adapt the step parameters based on the obstacle shape. Cubic polynomials

are the most prevalent [68], [69], [78], but seventh-order polynomial splines have also been employed [72], [73]. Other solutions include MPC [82], [159] or simple interpolation based on reference trajectories [130], [161], [162]. The movement constraints used for trajectory adaptation in the majority of EAGP solutions are expressed in terms of mathematical expressions that yield updated values for step length, height, and duration; although this approach seems reasonable when it comes to foothold selection (e.g., knowing the obstacle length can be enough to calculate the appropriate step length to cross it), the specific characteristics of the environment might also require a modification of the swing foot trajectory shape. For example, in [68] and [69], both foothold selection, step height, and trajectory shape are computed in an optimization fashion: the foothold is selected according to a sliding window approach that gives a score to each possible foothold based on physiological step length and obstacle presence, while the vertical peak of the swing foot trajectory is computed based on a stochastic iterative algorithm which aims to minimize kinematic (joint angles) and obstacle (euclidean distance) penalties during the motion. While this solution only optimizes the trajectory during planning, other solutions do the same in a dynamic fashion, allowing for real-time adaptation during execution: in [82], nonlinear MPC allows to compute a motion which satisfies both stability criteria of the CoM and obstacle avoidance, while in [73], the trajectory is recomputed starting from its current point whenever the angular position of the torso or the stair estimation output change their value significantly; in both of these cases, the foothold can change in real time, differently from all the other methods previously mentioned. Finally, it is worth highlighting that all the aforementioned EAGP solutions consider only LLE movement in the sagittal plane. Differently, in [164], a visual 2-D footstep planner for LLEs has been developed; a RealSense RGB-D camera (D435) is used to build a 3-D cost map of the surrounding environment, while a mixed reality device (Microsoft Hololens) is used to measure the user's intention in the form of gaze mapping and gesture recognition, in order to identify the desired goal of the walking path. Subsequently, a path planner based on Bezier curves is proposed to produce a 2-D trajectory on the horizontal plane, and a footstep planner considering multiple features (terrain, stepping, and tracking features) is employed to select the optimal footsteps sequence along the path planned beforehand. Moreover, thanks to mixed reality, the footstep sequence is displayed in real time, providing to the user with a visual feedback of the exoskeleton behavior.

VII. Discussion

This article summarizes recent advances in EAGP for LLEs, focusing on two core pillars: adaptive gait generation methodologies and environment understanding techniques. A consistent conceptual taxonomy remains absent in the literature; however, the proposed taxonomy is supported by related frameworks discussed in other works, such as "environment-oriented adaptive gait planning" [39] and "environmental information-based interactive control" [40].

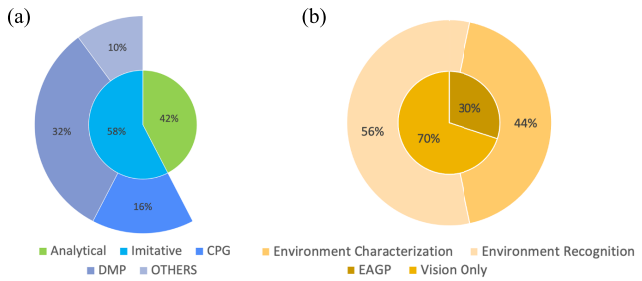


Fig. 6. (a) Adaptive gait generation method distribution over the review articles. (b) Environment understanding method distribution over the review articles.

The complete survey of the reviewed articles is summarized in the Supplementary Materials for adaptive gait generation (Table S-II), environment understanding (Table S-III), and EAGP (Table S-IV), highlighting the main attributes of each work.

A. Comparison of Adaptive Gait Generation Approaches

In-depth analysis of past and recent literature allowed us to identify two core approaches to the problem of adaptive gait generation in LLEs. Analytical methods, rooted in dynamic models such as the LIPM and hand-crafted trajectories formulation, surely represent the first approach for gait generation in both LLEs [31] and humanoid robots [165]. In fact, closed-form solutions (e.g., polynomial or sinusoidal trajectory parameterizations) ensure predictable behavior and low computational overhead, enabling real-time execution on embedded hardware [69], [82]. Real-time performance is crucial for semiautonomous LLEs, where the system must seamlessly adjust gait parameters at runtime, ensuring both safety and comfort for the user. However, these methods require manual tuning of model parameters to accommodate novel terrains or individual user biomechanics, limiting their generalization ability beyond structured scenarios. For instance, ZMP-based stair-ascending controllers provide stable ascent on uniform steps, but may fail under uneven or partially occluded treads without significant recalibration [166]. Thus, when also considering the adaptation of gait trajectories to the current environment, analytical methods are less used than imitative methods, as shown in Fig. 6(a). Indeed, imitative methods provide a more flexible description of human locomotion through a data-driven approach by learning gait trajectories from human demonstrations or through human interactions. Among these methods, DMPs excel at encoding both discrete and rhythmic movements via a canonical phase system and regression-based forcing functions, resulting in the most used method for adaptive gait generation in LLEs [Fig. 6(a)]. The strength of this method is highlighted in numerous works that use DMPs to adapt the gait to slopes [57], [58], [89], stairs [90], [91], [92], and step trajectory variations [95], [96] with minimal retuning. Nevertheless, since the stability of most DMP methods is based on the underlying spring–damper transformation system, abrupt environmental changes can induce transient instabilities [95]. Subsequent

evolutions of the original DMP framework (SDMPs, ProMPs, KMPs, and rhythmic DMPs) further increase generalization capabilities, but the increasing use of probabilistic elements reduces the guarantees on the accuracy of the final motion [106]. In parallel to DMP implementation, CPG-based controllers have been proposed to inherently generate a rhythmic locomotion robust to perturbations, and that can be easily modulated in frequency and phase in response to sensory feedback. In general, DMP and CPG methods share many similarities: in fact, CPGs can be implemented following the general (rhythmic) DMP framework proposed by Ijspeert et al. [85], as done in [112]. Different from analytical approaches that are commonly derived from the humanoid robots field, imitative approaches are particularly suited for exoskeleton application as they provide a probabilistic solution for encoding the statistical properties of the user's locomotion into the control loop of the wearable robot. In particular, the reliance of CPG methods on real-time torque or inertial input encourages seamless user–exoskeleton interaction, but requires careful design of feedback loops to avoid oscillatory divergence under sensor noise [113]. Indeed, as shown in Fig. 7, after the introduction of DMP first in 2018 [95] and CPG in 2020 [113], the number of works employing imitative solutions for adaptive gait planning showed an important increase with respect to analytical solutions. Despite their diffusion, the combination of CPGs with visual input has not yet been explored in LLEs. Nevertheless, the promising results of recent work using vision-guided CPG models for humanoid [167], and multilegged robots [168], [169] suggest that this could represent a viable solution for future research on EAGP for LLE applications. In our taxonomy, methods using analytical models for all joints are labeled analytical, while those including any imitative component are considered imitative. Nevertheless, hybrid frameworks that integrate analytical constraints with learned trajectories represent a promising middle ground. For example, embedding DMP-generated trajectories in conjunction with LIPM CoM modeling and the related stability criteria provides both adaptability and stability guarantees [57], [63]. Similarly, stiffness-modulated CPGs constrained by CoM dynamics can exploit bio-inspired flexibility while bounding CoM excursions [114], [170]. RL methods [79], [83] also represent an interesting middle ground between the two categories. Since their behavior is mainly influenced by reward function shaping, an appropriate design could result in physiologically constrained trajectories. Moreover, the very low inference time is a major feature for real-time applications.

B. Advances and Limitations in Environment Understanding

The use of computer vision is a relatively new addition to the field of LLEs. Unlike other reviews [21], [40], herein, we decided to classify the environment understanding methods according to the type of output provided to the different levels of the LLE controller. Environment recognition techniques provide a high-level discrete description of the surroundings that can be fed to the LLE mid-level controller, or combined with information from the high-level controller, to select between multiple precoded reference trajectories for different terrains. Among the various methods, CNN and hybrid edge

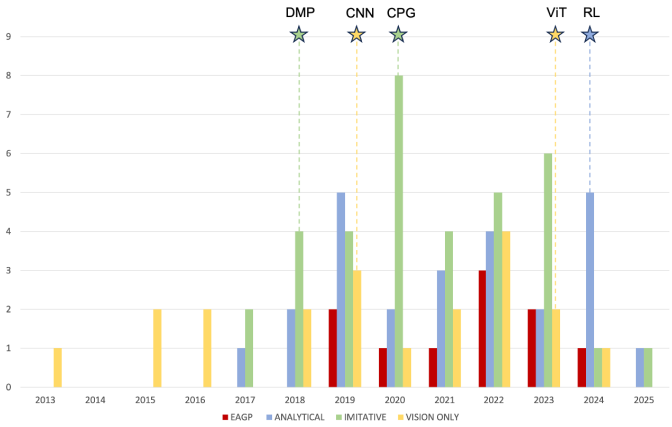


Fig. 7. Number of publications per year in relation to the proposed taxonomy categories. The algorithmic milestones are also highlighted.

detection pipelines showed the best performance, achieving rapid classification of walking contexts (e.g., stairs or slopes versus level ground) within 10–50 ms on embedded GPUs [144]. Nevertheless, current approaches to environment recognition still do not consider the transition between subsequent terrains, or present poor performance in these scenarios [143], [151]. To overcome this limitation, future work could consider the combination of these vision-based approaches with well-established methods in the literature that estimate the environment and locomotion mode from the internal state of the exoskeleton (e.g., through IMUs or footsole pressure sensors) [171]. Thanks to the great advancements of image and video classifications in computer vision, in particular through neural networks such as CNN in 2019 and transformers in 2023 [134], [143], [151], [152] (see Fig. 7), and the increasing availability of open-source datasets [139], [145], [146], [172], we found that the problem of environment recognition for LLEs is the most studied in the literature, as shown in Fig. 6(b). However, the lack of precise geometric data leads to conservative gait adaptations that reduce efficiency or fail on nonuniform terrains. For this reason, their use in complete EAGP solutions is scarce [Fig. 6(b)], with just one work applying it without the addition of environment characterization refinements [162]. In contrast, characterization-based methods using RGB-D or LiDAR sensors extract stairs dimensions [70], [72], [130], [149] or the inclination of a slope [133] with high accuracy. Approaches employing RANSAC plane fitting and region growing can estimate stair riser heights within 3% deviation but incur processing times of 1–4 s per frame, emphasizing the tradeoff between precision and latency [154]. These techniques can be found in almost every EAGP solution, but are often tailored to specific scenarios [69], [72], [130]. Emerging sensor fusion paradigms aim to bridge this gap: low-latency scene classification could trigger coarse gait adjustments (such as locomotion mode transitions) [143], while parallel depth-based characterization could fine-tune trajectories before execution [73]. Furthermore, the integration of SLAM-based mapping frameworks allows the memory of previously encountered obstacles beyond the sensor FOV, supporting anticipatory gait planning in dynamic environments

[133]. Nonetheless, robust fusion requires synchronized timestamps, calibrated extrinsic parameters, and outlier rejection strategies to mitigate drift and ensure safety [173]. When it comes to the sensors used in environment understanding, it is clear that RGB-D cameras provide the most comprehensive set of information: other ranged sensors (RGB cameras and laser range finders) have shown to be viable options, but since the output is less informative and structured, multiple scans and a significant amount of postprocessing is required to obtain a high-quality representation of the environment. The placement of such sensors must also be chosen carefully: head, chest, waist, lower limbs, and feet are viable positions, with the waist being the one that presents the most advantages [39]: the sensor input is the most stable, and the sensor can be rigidly connected to the LLE, synchronizing with its movement. Upper positions (head and chest) can increase the FOV of the sensor, but their movement is difficult to relate to the LLE, since the connection between the sensor and the LLE is not rigid [164]; on the other hand, lower positions (lower limbs and feet) can achieve higher precision when it comes to terrain analysis [72], but their FOV is very limited and they are subjected to wider movements with respect to the waist, decreasing the stability of the input [39].

C. EAGP System Integration: Current Challenges

Given the relatively new development of EAGP solutions for LLEs, the current state of the art is focused mainly on the implementation of standalone methodologies for adaptive gait generation and environment understanding, and only a limited number of works effectively integrated a full EAGP pipeline on a real exoskeleton device, as discussed in Section VI and shown in Fig. 6(b). Thus, incorporating EAGP modules into traditional hierarchical exoskeleton controllers remains a significant open problem. In this direction, based on the proposed literature review, we identified the following scientific and technological challenges.

1) *Sensor Synchronization and Latency Management:* Disparate sampling rates (e.g., inertial sensors at 1 kHz versus depth cameras at 30 Hz) pose the need for buffering and interpolation schemes to align data streams without introducing excessive delay.
2) *Safety*: The autonomy of the exoskeleton poses additional safety risks, particularly when the exoskeleton behaviors conflict with the user's intent. To this, arbitration logic should be introduced to reconcile user intent (e.g., manual commands) with environmental constraints, avoiding abrupt gait pattern switches that could destabilize the user.
3) *Device Generalization:* To maximize the impact of the newly developed EAGP algorithms, it is important to ensure the generalization to different types of exoskeleton devices, which could vary in their applications, power, and the number of actuated joints. Designing modular software architectures with clearly defined interfaces [e.g., robot operating system (ROS) action servers, real-time middleware, and on-board GPUs] can facilitate flexible experimentation and deployment across diverse hardware platforms.
4) *Computational Resource Allocation:* Wearable exoskeletons have limited onboard processing: when selecting algorithms, particular consideration should be given to the tradeoff between performance and computational cost.
5) *Ethical Considerations:* The use of exteroceptive sensors poses a critical problem related to data privacy, particularly with the use of cameras recording private spaces and other people [174]. The definitions of clear guidelines and rules on the appropriate use of these technologies must be identified by technical and ethical committees involving engineers, clinicians, and lawyers.

D. Limitations and Future Directions

Despite the growing field of EAGP for LLEs, current solutions at the state of the art present some common limitations that must be overcome to ensure the uptake of this technology outside laboratories. First, to provide a full description of human bipedal locomotion, gait trajectories should be designed for both the single support phase (i.e., when one of the legs is swinging forward) and the double support phase (when both feet are on the ground and the weight is transferred from one to the other). However, most of the solutions in the literature focus only on the single support phase, due to the relatively short duration of the double support phase (i.e., the single support phase accounts for around 80% of the gait cycle [62]). This gap could compromise an effective and seamless integration of user and exoskeleton movements during continuous locomotion. Additionally, a proper modeling of the double-support phase could lead to a more stable and safe gait execution [175], [176]. In this direction, rhythmic DMPs and CPGs have been shown to be promising in encoding the full gait cycle in trajectory generation with smooth transitions between multiple subsequent steps [104], [105], [117], [118], [119]. Another limitation of the state of the art is the fact that current EAGP solutions consider only the motion generated in the sagittal plane. The reason is twofold: on the one hand, the locomotion task is predominantly executed along this plane; on the other hand, many exoskeletal systems are not equipped with the necessary actuated joints to perform active turning. Generally, turning is performed manually by the user through the use of external supports (e.g., crutches), incrementally twisting the upper body and turning in place [177]. Only a few exoskeleton systems also include the necessary DoF for active turning [56], [178], and we identified only one work that proposes an EAGP method that considers gait generation outside the sole sagittal plane [164]. This limitation is more related to the limited number of exoskeleton platforms capable of performing active turning. As a result, its practical relevance is likely limited until safe and effective methods for active turning are achieved. An additional limitation of current EAGP solutions is the lack of generalization to different environmental scenarios. As can be seen in Fig. 8, there is a huge gap between solutions tailored for level ground walking with respect to the other environments. Stairs are the second most investigated scenario in both environment understanding and adaptive gait generation studies. However, the number of works that effectively tested

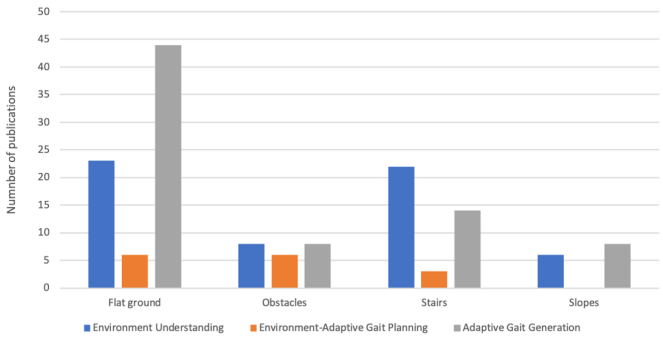


Fig. 8. Number of publications for each notable environmental scenario in the context of adaptive gait generation, environment understanding, and EAGP.

a vision-guided exoskeleton in stairs up/down experiments is scarce. Recent solutions have increasingly explored obstacle avoidance for LLEs, with successful evaluations on real robotic platforms [69], [78], [82]. In contrast, there is a lack of EAGP solutions that address slope scenarios. In the end, we identified only one adaptive gait generation method that explicitly addresses all the main indoor environmental occurrences (flat ground, stairs, slopes, and the presence of obstacles) [19], and there is no ongoing research on outdoor gait planning for LLEs. This gap can likely be attributed to the structured geometry of indoor scenarios, which greatly aids computation. Outdoor environments, with their irregular terrain and variable light conditions, might cause incorrect vision output or unreliable trajectory generation, particularly affecting CoM balance. In general, despite increasing evidence on the importance of relating visual perception and movement planning in wearable robotics [179], the link of computer vision with gait generation in LLEs is still in its infancy. Finally, the field of EAGP is characterized by fragmented evaluation benchmarks, reliance on small subject cohorts (often non-disabled volunteers), and limited reporting of negative or inconclusive results. Simulation-based validations, while valuable for algorithm development, may not capture real-world variability such as uneven terrain, sensor occlusion, or user fatigue. Furthermore, the lack of standard taxonomies for EAGP components and performance metrics hinders direct methods comparison. In this article, a taxonomy of different adaptive gait generation and environment understanding methods has been introduced although a benchmark to compare them is still missing. Since this article aims to provide a comprehensive overview of the computational methods used to achieve EAGP, the formulation of a standardized evaluation benchmark is beyond its scope. Nonetheless, human gait datasets [180] represent a strong reference in terms of data collection protocols, collected data requirements, and evaluation metrics. In this direction, events such as the Cybathlon [181] provide a unique opportunity to qualitatively and quantitatively evaluate the performance of EAGP in LLEs in complex scenarios simulating real-life conditions.

VIII. Conclusion

In this article, a review of EAGP solutions for LLEs is presented. As main contributions, this article provides a refined definition of EAGP integrated within the standard LLE control framework. Moreover, a novel taxonomy for adaptive gait generation and environment understanding methods in exoskeleton applications is introduced to categorize the extensive survey of the existing literature. In addition, to aid the reader in comprehending the breadth and depth of the reviewed literature, detailed tables summarizing the reviewed articles and their classification according to the proposed taxonomy are provided in the Supplementary Materials, offering a valuable resource for further exploration and analysis. We believe that this article serves as a critical resource for driving future advancements in the design and deployment of semiautonomous LLEs, ultimately enhancing their efficacy and applicability in everyday life.

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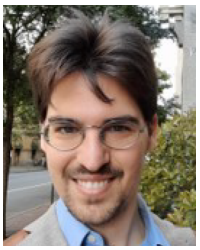
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Mr. Trombin also participated in the Cybathlon 2024 BCI Race as a member of the WHi Team, which earned the second place in the international competition.



Stefano Tortora (Member, IEEE) received the M.Sc. degree in biomedical engineering from the Politecnico di Milano, Milan, Italy, in 2017, and the Ph.D. degree in information engineering from the University of Padua, Padua, Italy, in 2021.

Since 2023, he has been an Assistant Professor with the Department of Information Engineering, University of Padua. His research has been passionately dedicated to improving the interaction between users and robotic devices by creating a new paradigm in human–machine interfacing (HMI) fusing multimodal information coming from wearable sensors and robots' sensors for the development of intelligent assistive devices and exoskeletons aimed at enhancing the mobility of people with motor impairments.

Dr. Tortora won the Cybathlon BCI Series and the Cybathlon Global editions in 2019 and 2020 as a member of the WHi Team, and the silver and bronze medals at the Cybathlon 2024 Event as a Leader of the WHi Team and the WHi Students Team in 2024. He also received the Cybathlon 2024 BCI Jury Award for the scientific innovation of his research, the Best Paper Award at the 2023 IEEE International Conference on Metrology for Extended Reality, Artificial Intelligence and Neural Engineering, and a nomination to the Best Student Paper Award at the 2020 IEEE International Conference on Systems, Man, and Cybernetics.



Francesco Bettella received the M.Sc. degree in mechanical engineering and the Ph.D. degree in neuroscience from the University of Padua, Padua, Italy, in 2016 and 2025, respectively.

Since 2016, he has been conducting many research projects focused on assistive technologies and robotic devices for rehabilitation with the Department of Industrial Engineering and the Padova Neuroscience Center, University of Padua.

Dr. Bettella won the Cybathlon BCI Series and the Cybathlon Global editions in 2019 and 2020 as a Pilot of the WHi Team, and the Silver Medal at the Cybathlon 2024 Event as a Pilot of the WHi Team in 2024.



Alessandra Del Felice received the Ph.D. degree in neuroscience from the University of Verona, Verona, Italy, in 2013.

She is an Academic Clinician of Neurorehabilitation with the University Hospital Padova, Padova, Italy. Her basic training has been as a neurologist, with a special interest in neurophysiology and neurorehabilitation. Her broad network of national and international collaborations and partnerships with small and medium enterprises (SMEs) ensures a robust knowledge exchange and provides access to cutting-edge technological and computational resources to achieve these objectives. She directs the Laboratory of Rehabilitation, Neuroscience, and Technologies (ReNeT Lab) and leads the "Robotics and Neuroscience" Working Group, Padova Neuroscience Center, University of Padua, Padua, Italy. Her current research area allows her to draw on her experience and skills in neurophysiology to improve methodology and outcome measures in this important area of rehabilitation.

Dr. Del Felice is a Co-Leader of the Working Group on Neurophysiology in Neurorehabilitation and a member of the Working Group on Stroke of Italian Society of Physical and Rehabilitation Medicine (SIMFER). She is also an Active Member of the Stroke Recovery and Rehabilitation Roundtable (SRRR) and a Founding Member of European Academic Organization for Physical and Rehabilitation Medicine (UniPMR).



Emanuele Menegatti (Senior Member, IEEE) was the European Coordinator of the FP7 FoF-EU Project "Thermobot" and a local principal investigator of other five European projects. He is a Full Professor with the Department of Information Engineering, University of Padua, Padua, Italy. In particular, he is working on Neurorobotics and Collaborative Robotics for Industry 4.0. He is a WP Leader of a PNRR Italian Project on Wearable Robots (i.e., exoskeletons) controlled by electromyography (EMG) and EEG signals for the elderly. He is teaching the courses "Intelligent Robotics" and "Data Structures and Programming," has supervised 21 Ph.D. students, 13 postdoctorals, more than 60 master's theses, and more than 15 bachelor's theses, and was a Co-Supervisor of two Japanese Ph.D. students. He is a Founding Partner of two startups, IT+Robotics srl, Padova, Italy, and EXiMotion srl, Padova. He has authored more than 50 articles in international journals and more than 120 publications in international conferences, and has co-authored five patents. His research interests are in the field of robot perception.

Dr. Menegatti is an Associate Editor of *Robotics and Autonomous Systems* (Elsevier), *Robotica* (Cambridge Press), *IPSJ Transactions on Computer Vision and Applications* (Springer), *Frontiers in Neurorobotics* (Frontiers), and *International Journal of Advanced Robotic Systems* (Sage). He has served as an Associate Editor for the IEEE ICRA Conference, the IEEE IROS Conference, and the IJCAI Conference. Since 2020, he has been the President of the International Intelligent Autonomous System Society (IAS-Soc). He also served as a Project Reviewer for European Commission in FP7 and H2020.



Luca Tonin (Senior Member, IEEE) received the Ph.D. degree in robotics from the École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland, in 2013.

Since 2024, he has been an Associate Professor with the Department of Information Engineering, University of Padua, Padua, Italy. His research interests include brain–machine interfaces and their application to robotic devices.

Dr. Tonin won the Cybathlon Event in 2019 as a Co-Leader of the BrainTweakers Team, the Cybathlon BCI Series and the Cybathlon Global editions in 2019 and 2020 as a Leader of the WHi Team, and the silver and bronze medals at the Cybathlon 2024 Event as a Leader of the WHi Team and the WHi Students Team in 2024.

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